

Instructor: *Profs. Hershfield and Takano*

PHYSICS DEPARTMENT

PHY 2049

Final Exam

May 2, 2023

Name (print, last first): _____ Signature: _____

On my honor, I have neither given nor received unauthorized aid on this examination.

TA: _____ Discussion period and day: _____

BEFORE YOU START

- (1) Print and sign your name on this QUESTION sheet (in the space provided above).
- (2) Your TEST NUMBER is the 5-digit number at the top of this QUESTION SHEET.
- (3) Code your 5-digit test number onto lines 76 – 80 of your ANSWER SHEET (scan sheet).
- (4) Code your name and UFID on your ANSWER SHEET.
- (5) For each question, blacken the circle of your intended answer completely on the answer sheet. DARKEN CIRCLES COMPLETELY, using a No. 2 pencil or blue or black ink. Do not make any stray marks on the answer sheet.
- (6) You can do scratch work on this question sheet. Additional scratch paper is available if you want it.
- (7) For each question, choose the best answer. Many answers are rounded. Choose the most correct answer. All questions carry equal credit. There is no penalty for guessing.
- (8) You may use a simple electronic calculator - not a cell phone. No books, notes, or other resources are permitted.
- (9) You may separate the included formula sheets and question pages if you wish. There is no need to reattach or staple them later.

WHEN YOU FINISH

- (1) Check that you have coded your name, UFID and test number on your answer sheet (scan sheet). Check that your name and signature are on this question sheet.
- (2) Turn in your answer sheet (scan sheet) AND question sheet. Please keep your scratch paper.
- (3) No credit will be given unless you turn in both the answer sheet and the question sheet with your name on each.

Formulas – page 1

Constants:

$$\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A} \quad \epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2.$$

$$e = 1.6 \times 10^{-19} \text{ C} \quad k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2.$$

$$c = 3.0 \times 10^8 \text{ m/s}$$

kilo (k) = 10^3 ; centi (c) = 10^{-2} ; milli (m) = 10^{-3} ; micro (μ) = 10^{-6} ; nano (n) = 10^{-9} ; pico (p) = 10^{-12}

Coulomb's Law:
$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2}$$

Electric field:

$$\vec{F} = q\vec{E} \quad E = \frac{qz}{4\pi\epsilon_0(z^2 + R^2)^{3/2}} \quad (\text{charged ring})$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2} \quad (\text{point charge}) \quad E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}}\right) \quad (\text{charged disk})$$

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \quad (\text{continuous}) \quad E = \frac{\sigma}{2\epsilon_0} \quad (\text{infinite sheet})$$

Dipoles:
$$p = qd \quad \tau = pE \sin \theta \quad U = -pE \cos \theta$$

Gauss' Law:
$$\Phi = \int \vec{E} \cdot d\vec{A} = (E \cos \theta)A \quad E = \frac{\sigma}{\epsilon_0} \quad (\text{conducting surface})$$

$$\epsilon_0\Phi = \epsilon_0 \oint \vec{E} \cdot d\vec{A} = q_{\text{enc}} \quad E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (\text{line of charge})$$

Electric potential:
$$U = qV \quad V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad (\text{point charge})$$

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s} \quad dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} \quad (\text{continuous})$$

$$E_x = -\frac{\partial V}{\partial x}; \quad E_y = -\frac{\partial V}{\partial y}; \quad E_z = -\frac{\partial V}{\partial z}. \quad U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} \quad (\text{two-particle system})$$

Capacitors:
$$q = CV \quad C = \frac{\epsilon_0 A}{d} \quad (\text{parallel plate})$$

$$C_{\text{eq}} = \sum_{j=1}^n C_j \quad (n \text{ capacitors in parallel}) \quad C = 2\pi\epsilon_0 \frac{L}{\ln(b/a)} \quad (\text{cylindrical})$$

$$\frac{1}{C_{\text{eq}}} = \sum_{j=1}^n \frac{1}{C_j} \quad (n \text{ capacitors in series}) \quad C = 4\pi\epsilon_0 \frac{ab}{b-a} \quad (\text{spherical})$$

$$U_E = \frac{q^2}{2C} = \frac{1}{2} CV^2, \quad C = 4\pi\epsilon_0 R \quad (\text{isolated sphere})$$

$$u_E = \frac{1}{2} \epsilon_0 E^2. \quad C = \kappa \frac{\epsilon_0 A}{d} \quad (\text{parallel plate with dielectric})$$

Formulas – page 2

Current:	$i = \frac{dq}{dt},$	$i = \int \vec{J} \cdot d\vec{A},$	$\vec{J} = (ne)\vec{v}_d,$	
Resistors:	$R = \frac{V}{i},$	$v_R = iR.$	$R = \rho \frac{L}{A},$	$\rho = \frac{1}{\sigma} = \frac{E}{J},$
	$\frac{1}{R_{eq}} = \sum_{j=1}^n \frac{1}{R_j}$	$(n \text{ resistances in parallel}).$		$R_{eq} = \sum_{j=1}^n R_j$ $(n \text{ resistances in series}).$
Inductors:	$L = \frac{N\Phi_B}{i}$	$\mathcal{E}_L = -L \frac{di}{dt}.$	$\frac{L}{l} = \mu_0 n^2 A$	$(\text{solenoid}).$
	Inductors add in series and parallel with the same rules as resistors.			$\mathcal{E}_2 = -M \frac{di_1}{dt}$
	$U_B = \frac{Li^2}{2},$	$u_B = \frac{B^2}{2\mu_0}$	$\mathcal{E}_1 = -M \frac{di_2}{dt},$	
RC circuit:	$q = C\mathcal{E}(1 - e^{-t/\tau_C}).$	(charging)	$q = q_0 e^{-t/\tau_C}.$	(discharging)
			$\tau_C = RC.$	
RL circuit:	$i = \frac{\mathcal{E}}{R}(1 - e^{-t/\tau_L})$	$(\text{rise of current}).$	$i = i_0 e^{-t/\tau_L}$	$(\text{decay of current}).$
			$\tau_L = \frac{L}{R}$	
LC circuit:	$q = Q \cos(\omega t + \phi)$	$(\text{charge}),$	$i = \frac{dq}{dt} = -\omega Q \sin(\omega t + \phi)$	$(\text{current}).$
			$\omega = \frac{1}{\sqrt{LC}}.$	
RLC circuit: (not driven)	$q = Q e^{-Rt/2L} \cos(\omega' t + \phi),$		$\omega' = \sqrt{\omega^2 - (R/2L)^2}.$	
Driven circuits:	$\mathcal{E} = \mathcal{E}_m \sin \omega_d t.$			
One circuit element:	R		$i_R = \frac{v_R}{R} = \frac{V_R}{R} \sin \omega_d t.$	
	$X_C = \frac{1}{\omega_d C}$	$(\text{capacitive reactance}).$	$i_C = \left(\frac{V_C}{X_C}\right) \sin(\omega_d t + 90^\circ).$	
	$X_L = \omega_d L$	$(\text{inductive reactance}).$	$i_L = \left(\frac{V_L}{X_L}\right) \sin(\omega_d t - 90^\circ).$	
Series RLC:	$Z = \sqrt{R^2 + (X_L - X_C)^2}$	(impedance)	$i = I \sin(\omega_d t - \phi),$	
	$\tan \phi = \frac{X_L - X_C}{R}$	$(\text{phase constant}).$	$I = \frac{\mathcal{E}_m}{\sqrt{R^2 + (X_L - X_C)^2}}.$	
Power:	$I_{rms} = \frac{I}{\sqrt{2}}$	$\mathcal{E}_{rms} = \frac{\mathcal{E}_m}{\sqrt{2}}$	$P_{avg} = \mathcal{E}_{rms} I_{rms} \cos \phi$	
Transformers:	$V_s = V_p \frac{N_s}{N_p}$	$I_s = I_p \frac{N_p}{N_s}$	$R_{eq} = \left(\frac{N_p}{N_s}\right)^2 R,$	

Formulas – page 3

Forces: $\vec{F} = q\vec{E}$, $\vec{F}_B = q\vec{v} \times \vec{B}$; $F_B = |q|vB \sin \phi$, $\vec{F}_B = i\vec{L} \times \vec{B}$. $F_{ba} = \frac{\mu_0 Li_a i_b}{2\pi d}$

Circular motion: $|q|vB = \frac{mv^2}{r}$, $r = \frac{mv}{|q|B}$. $f = \frac{\omega}{2\pi} = \frac{1}{T} = \frac{|q|B}{2\pi m}$. Hall effect: $n = \frac{Bi}{Vle}$,

Magnetic moment: $\mu = NiA$ $\vec{\tau} = \vec{\mu} \times \vec{B}$. $\tau = NiAB \sin \theta$, $U(\theta) = -\vec{\mu} \cdot \vec{B}$.

Magnetic field: $d\vec{B} = \frac{\mu_0}{4\pi} \frac{i d\vec{s} \times \hat{r}}{r^2}$ (Biot–Savart law) $B = \frac{\mu_0 i}{2\pi R}$ (long straight wire)

$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{\text{enc}}$ (Ampere's law) $B = \frac{\mu_0 i \phi}{4\pi R}$ (at center of circular arc)

$B = \frac{\mu_0 i (2\pi)}{4\pi R} = \frac{\mu_0 i}{2R}$ (at center of full circle)

$B = \mu_0 in$ (ideal solenoid), $B = \frac{\mu_0 i N}{2\pi} \frac{1}{r}$ (toroid)

Faraday's law: $\mathcal{E} = \oint \vec{E} \cdot d\vec{s}$. $\mathcal{E} = -\frac{d\Phi_B}{dt}$ $\Phi_B = \int \vec{B} \cdot d\vec{A}$

Maxwell's eqs.: Gauss' law for electricity $\oint \vec{E} \cdot d\vec{A} = q_{\text{enc}}/\epsilon_0$

Gauss' law for magnetism $\oint \vec{B} \cdot d\vec{A} = 0$

Faraday's law $\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$

Ampere–Maxwell law $\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} + \mu_0 i_{\text{enc}}$ $i_d = \epsilon_0 \frac{d\Phi_E}{dt}$.

Polarizers: $I = \frac{1}{2} I_0$. $I = I_0 \cos^2 \theta$.

Refraction: $n_2 \sin \theta_2 = n_1 \sin \theta_1$, $\theta_c = \sin^{-1} \frac{n_2}{n_1}$ (critical angle).

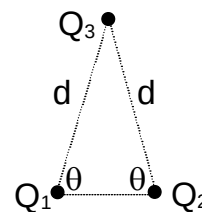
Mirrors: $i = -p$ (plane mirror). $\frac{1}{p} + \frac{1}{i} = \frac{1}{f} = \frac{2}{r}$. $m = -\frac{i}{p}$.

Lenses: $\frac{1}{f} = \frac{1}{p} + \frac{1}{i}$ (thin lens), $m = -\frac{i}{p}$ (magnification),

Two slits: $d \sin \theta = m\lambda$, for $m = 0, 1, 2, \dots$ (maxima—bright fringes), $d \sin \theta = (m + \frac{1}{2})\lambda$, for $m = 0, 1, 2, \dots$ (minima—dark fringes),

Single slit: $a \sin \theta = m\lambda$, for $m = 1, 2, 3, \dots$ (minima). $I(\theta) = I_m \left(\frac{\sin \alpha}{\alpha} \right)^2$, where $\alpha = \frac{\pi a}{\lambda} \sin \theta$

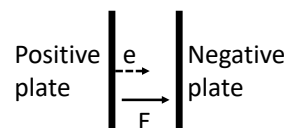
1. Two point charges of charge $Q_1 = 2\mu\text{C}$ and $Q_2 = -2\mu\text{C}$ are placed at the base of the triangle shown in the figure with $\theta = 70^\circ$ and $d = 5\text{cm}$. What is the magnitude of the force on a third charge placed at the top of the triangle with charge $Q_3 = 2\mu\text{C}$?



- (1) 9.84 N (2) 4.92 N (3) 28.8 N (4) 14.4 N (5) 20.4 N

Solution: Q_3 is repelled from Q_1 and attracted to Q_2 . Drawing a vector diagram, the net force on Q_3 is in the x-direction and equal to $2 \cos(70^\circ) k(2\mu\text{C})^2 / (5\text{cm})^2 = \mathbf{9.84\text{ N}}$.

2. An electron is shot with initial velocity $8.4 \times 10^5\text{ m/s}$ from a positive plate towards a negative plate of the opposite charge density. The voltage between the plates is 5 V, and the separation between the plates is 2 cm. What is the maximum distance that the electron gets from the positive plate? The mass of an electron is $9.1 \times 10^{-31}\text{ kg}$.



- (1) 0.8 cm (2) 0.4 cm (3) 1.2 cm (4) 1.6 cm (5) 2.0 cm

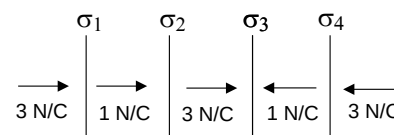
Solution: The magnitude of the electric field between the plates is $(5\text{V})/(2\text{cm}) = 250\text{V/m}$. The electron turns around when its velocity is zero. The magnitude of the acceleration is $a = eE/m_e$.

$$v = 0 = v_o - at \rightarrow t = v_o/a$$

The position at that time is

$$x = v_o t - 0.5at^2 = 0.5 v_o^2/a = 0.5 m_e v_o^2 / eE = 0.008\text{m} = \mathbf{0.8\text{cm}}$$

3. The numerical magnitude and direction (arrows) of the electric field due to four infinite parallel planes of charge with uniform charge densities σ_1 , σ_2 , σ_3 , and σ_4 are shown in the figure. What is the sum of the surface charge densities for planes 2, 3, and 4: $\sigma_2 + \sigma_3 + \sigma_4$?



- (1) $-35.4 \times 10^{-12}\text{ C/m}^2$
 (2) $+35.4 \times 10^{-12}\text{ C/m}^2$
 (3) $-17.7 \times 10^{-12}\text{ C/m}^2$
 (4) $+17.7 \times 10^{-12}\text{ C/m}^2$
 (5) $-8.85 \times 10^{-12}\text{ C/m}^2$

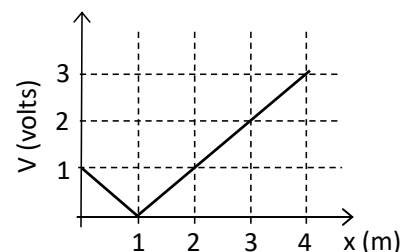
Solution: Use a Gaussian surface in the shape of a cylinder that goes through planes 2, 3, and 4. On the right hand side the flux is $(-3\text{N/C})A$, and on the left hand side the flux is $(-1\text{N/C})A$. Here A is the area of one of the ends of the cylinder. The negative sign is because the electric field is going into the surface. The sum of these is $(-4\text{N/C})A$ and equal to $(\sigma_2 + \sigma_3 + \sigma_4)A/\epsilon_o$ so that $(\sigma_2 + \sigma_3 + \sigma_4) = (-4\text{N/C})\epsilon_o = \mathbf{-35.4 \times 10^{-12}\text{C/m}^2}$.

4. The electric potential in a region of space is given by $V(x, y, z) = A(x^2y + z^3)$, where x , y , and z are measured in meters and $A = 2\text{ kV/m}^3$. What is the y -component of the electric field at $(x, y, z) = (1, 2, 3)$?

- (1) -2 kV/m (2) 2 kV/m (3) -8 kV/m (4) 8 kV/m (5) 54 kV/m

Solution: The y -component of the electric field is $-\partial V/\partial y = -Ax^2 = \mathbf{-2\text{kV/m}}$.

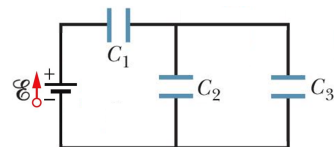
5. A charged particle of charge $2C$ and mass 0.5kg moves in the one dimensional electric potential shown in the figure. At $t = 0$ the particle has velocity $+2\text{m/s}$ and is at $x = 0\text{m}$. What is the maximum position (x) that the particle will reach?



- (1) 2.5 m (2) 2 m (3) 3 m (4) 3.5 m (5) 4 m

Solution: The initial kinetic energy is $0.5mv^2 = 1J$. The initial potential energy is $qV = 2J$. Thus, the total initial energy is $3J$. Finally, the kinetic energy is zero so the potential energy is $3J = qV = 2V$, which implies that $V = 1.5$. This occurs at $x = 2.5\text{m}$.

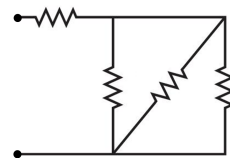
6. In the figure, $V = 12\text{ V}$, $C_1 = 8\text{ }\mu\text{F}$, $C_2 = 1\text{ }\mu\text{F}$, and $C_3 = 3\text{ }\mu\text{F}$. What is the voltage across capacitor C_3 , in volts?



- (1) 8 (2) 12 (3) 10 (4) 6 (5) 3

Solution: C_2 and C_3 are in parallel forming an effective capacitance of $4\mu F$. The effective capacitance of the entire circuit is $8/3\mu F$. The charge on C_1 and the effective $(C_2 + C_3)$ capacitor is $12V/(8/3\mu F) = 32\mu C$. The voltage across $C_2 + C_3$ is $32\mu C/4\mu F = 8V$. C_2 and C_3 have the same voltage because they are in parallel.

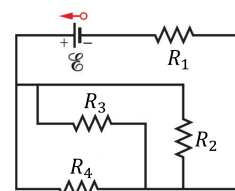
7. In the figure, all resistances are $300\text{ }\Omega$. What is the equivalent resistance of this combination, in ohms?



- (1) 400 (2) 1,200 (3) 800 (4) 600 (5) 300

Solution: The three 300Ω resistors on the right are in parallel and have effective resistance 100Ω . These are in series with the left 300Ω resistor, producing a total resistance of **400Ω** .

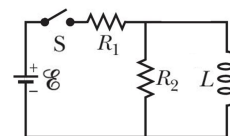
8. In the figure, $\mathcal{E} = 12\text{ V}$, $R_1 = R_2 = 100\text{ }\Omega$, and $R_3 = R_4 = 200\text{ }\Omega$. What is the current through resistor R_2 , in mA?



- (1) 40 (2) 70 (3) 60 (4) 30 (5) 10

Solution: R_2 , R_3 , and R_4 are in parallel with effective resistance 50Ω . This makes the total resistance 150Ω and the current through R_1 is $12V/150\Omega = 0.08A$. The voltage across the three resistors in parallel is $(50\Omega)(0.08A) = 4V$. The current through R_2 is $4V/R_2 = 0.04A = \mathbf{40mA}$.

9. In the figure, $\mathcal{E} = 3.0\text{ V}$, $R_1 = R_2 = 10\text{ }\Omega$, and $L = 20\text{ mH}$. What is the current (in A) through resistor R_1 long after the switch is closed?



- (1) 0.30 (2) 0.15 (3) 0 (4) 0.07 (5) 0.23

Solution: A long time after the switch is closed the voltage across the inductor is $0V$. This means that all the voltage is dropped across R_1 . The current is $3V/R_1 = \mathbf{0.3A}$.

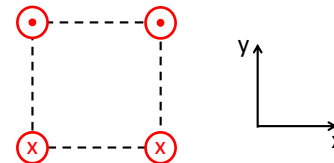
10. A negatively charged particle has a velocity in the plane of the page shown in the figure. Which of the following magnetic field directions would produce the force, F , on the particle that is shown in the figure?



- (1) out of the page (2) into the page (3) right (4) left (5) parallel to F

Solution: Because the particle is negatively charged and $\vec{F} = q\vec{v} \times \vec{B}$, the magnetic field is **out of the page**.

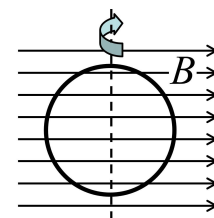
11. Four wires at the corners of a square have currents flowing perpendicular to the page with the directions shown in the figure. The magnitude of the currents is the same. What is the direction of the magnetic field at the center of the square?



- (1) $+\hat{i}$ (2) $-\hat{i}$ (3) $+\hat{j}$ (4) $-\hat{j}$ (5) $(\hat{i} + \hat{j})/\sqrt{2}$

Solution: The upper left wire produces a field pointing northeast, the upper right wire produces a field pointing southeast, the lower left wire produces a field pointing southeast, and the lower right wire produces a field pointing northeast. All of the fields have the same magnitude. Adding these magnetic fields as vectors produces a field pointing east or $\hat{i}$.

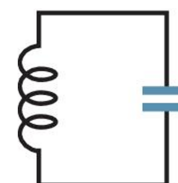
12. A circular loop of radius 10 cm, made of a conductor, is placed in a uniform horizontal magnetic field $B = 1.0$ T, as shown. The loop is rotated 60 times a second at a constant angular velocity about a vertical shaft (dashed line) that runs through the center. What is the amplitude of the emf induced in the loop, in volts?



- (1) 11.8 (2) 10.4 (3) 9.6 (4) 8.7 (5) 7.3

Solution: The magnetic flux as a function of time is $\Phi_B = B\pi r^2 \sin(\omega t)$ with $r = 0.1$ m and $\omega = 2\pi 60$ Hz. The induced emf is $-d\Phi_B/dt = -B\pi r^2 \omega \cos(\omega t)$, which has amplitude $B\pi r^2 \omega = \mathbf{11.8V}$.

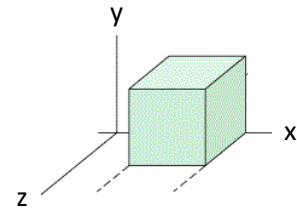
13. A $30 \mu\text{F}$ capacitor, with an initial charge of 1.0 mC , is connected to an inductor at time $t = 0$. In the ensuing oscillation, the maximum current in the circuit is 2.0 A. What is the inductance of the inductor, in mH?



- (1) 8.3 (2) 6.2 (3) 10.6 (4) 12.7 (5) 14.9

Solution: The maximum energy stored in the capacitor is equal to the maximum energy stored in the inductor because energy is conserved and both maxima occur when the other energy is zero. $0.5 q_m^2/C = 0.5 L i_m^2$ implies that $L = (q_m/i_m)^2/C = \mathbf{8.3mH}$.

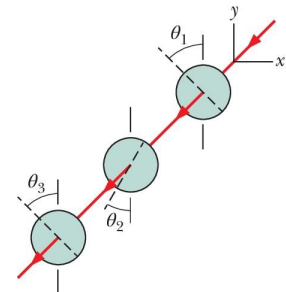
14. An electric field given by $\vec{E} = 10\hat{i} + 5(y^2 + 6)\hat{j}$ pierces a Gaussian cube of edge length 0.5m and positioned as shown in the figure. The magnitude of E is in Newtons per Coulomb and the position is in meters. What is the net electric flux through the cube?



- (1) $+0.312N \cdot m^2/C$ (2) $-0.312N \cdot m^2/C$ (3) $+1.25N \cdot m^2/C$ (4) $-1.25N \cdot m^2/C$ (5) $0N \cdot m^2/C$

Solution: The flux through the surfaces perpendicular to z-direction is zero because $E_z = 0$. The fluxes cancel for the x-surfaces because the electric field is constant. The flux through the top surface is $5(0.5^2 + 6)0.5^2$, and the flux through the bottom surface $-5(6)0.5^2$ with the minus coming because it is directed inwards. Thus, the net flux is $5(0.5^2)(0.5^2) = \mathbf{0.312N \cdot m^2/C}$.

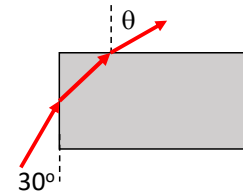
15. Initially unpolarized light is sent into a system of three polarizing sheets whose polarizing directions make angles of $\theta_1 = 0^\circ$, $\theta_2 = 30^\circ$, and $\theta_3 = 45^\circ$ with the direction of the y axis. What fraction of the initial intensity is transmitted by the system?



- (1) 0.0251 (2) 0.0625 (3) 0.1875 (4) 0.0084 (5) 0.3499

Solution: The angle between polarizers 1 and 2 is 30° , and the angle between polarizers 2 and 3 is $30 + 45 = 75^\circ$. Thus, the final intensity is $0.5I_o \cos^2(30) \cos^2(75) = \mathbf{0.0251}$.

16. Light is incident from air onto a rectangular slab material with index of refraction 1.2 with the angles shown in the figure. What is the angle θ that the light leaves the slab?



- (1) 56° (2) 45° (3) 67° (4) 78° (5) no light gets through

Solution: The angles in Snell's law are measured relative to the normal. Thus, the incident angle here is 60° . The transmitted angle is then $\sin^{-1}(1 \sin(60)/1.2) = 46.194^\circ$. After some trigonometry for right triangles, the angle of incidence at the second interface is $90 - 46.194 = 43.806^\circ$. The final angle of the transmitted light is $\sin^{-1}(1.2 \sin(43.806)/1) = 56.167^\circ \approx \mathbf{56^\circ}$.

17. A small object is located on the central axis of a spherical concave mirror, whose radius of curvature is 50 cm. If the object is a distance 10 cm from the mirror, what is the magnification?

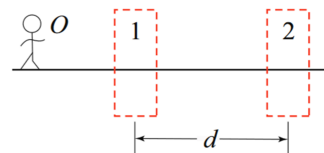
- (1) 1.7 (2) 2.5 (3) 3.4 (4) 4.1 (5) 5.0

Solution: Use

$$\frac{1}{p} + \frac{1}{i} = \frac{1}{f} = \frac{2}{r}$$

with $r = +50\text{cm}$ and $p = 10\text{cm}$ to find $i = 16.7\text{cm}$. The magnification is $m = -i/p = 1.67 \approx \mathbf{1.7}$.

18. In the figure, O (the object) stands on the common central axis of two thin, converging lenses, which are mounted in the boxed regions 1 and 2. The focal lengths of the lenses are both 10 cm. The object is 20 cm to the left of lens 1, which is a distance $d = 25$ cm from lens 2. What is the distance of the final image from lens 2, in cm? (The value is positive if the image is to the right of the lens, and negative if the image is to the left of the lens.)



- (1) -10 (2) -15 (3) 5 (4) 20 (5) 25

Solution: For the first lens $p_1 = 20\text{cm}$ and $f_1 = 10\text{cm}$ so $i_1 = 20\text{cm}$. Because $d = 25\text{cm}$, the distance to the second lens is $p_2 = 5\text{cm}$. This implies that $i_2 = -10\text{cm}$.

19. Sunlight is used in a double-slit interference experiment. The fourth-order maximum for a wavelength of 600 nm occurs at an angle of $\theta = 90^\circ$. Thus, it is on the verge of being eliminated from the pattern because θ cannot exceed 90° . To eliminate all of the visible light (wavelengths ranging from 400 nm to 700 nm) in the fourth-order maximum, what least change in the slit separation is needed, in nanometers?

- (1) 800 (2) 700 (3) 600 (4) 500 (5) 400

Solution: Using the double slit interference relation $d\sin(\theta) = m\lambda$ with $\theta = 90^\circ$, $m = 4$, and $\lambda = 600\text{nm}$, we see that $d = 4\lambda = 2400\text{nm}$. Decreasing d means that more wavelengths are eliminated. To eliminate $\lambda = 400\text{nm}$, one must have $d = 4\lambda = 1600\text{nm}$ for a decrease in slit separation of **800 nm**.

20. The distance between the first and fifth minima of a single-slit diffraction pattern is 5.0 mm with the screen 50 cm away from the slit, when light of wavelength 600 nm is used. What is the angle θ of the *third* diffraction minimum?

- (1) 0.43° (2) 0.74° (3) 0.92° (4) 1.27° (5) 1.86°

Solution: For small angles the single slit diffraction relation $a\sin(\theta) = m\lambda$ becomes $a\theta = m\lambda$ with θ measured in radians. The difference in angle between the first and fifth minima is $a\Delta\theta = 4\lambda$. Again using the small angle approximation, where $\tan(\theta) \approx \theta$ as well, $\Delta\theta = 5\text{mm}/50\text{cm} = 0.01$ in radians. For the third order diffraction minimum $a\theta = 3\lambda$, which implies that $\theta/\Delta\theta = 3/4$ and $\theta = 0.0075$ radians = **0.43°** .

Scratch paper

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